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Department: Information Technology

Bachelor of Information and Communications Technology (BINCT1) & Advanced Diploma in Information and Communications Technology (ADICT1)

MODULE / SUBJECT CODE: PBDV301 & PBDE401

Project 1 – Web-Based Application Project Due Date – 23 March 2026

GROUP NAME: Code Crafters

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Project Proposal

Purpose

Student-led workshops are currently informal, unorganized, and not tracked, resulting in few attendance and limited feedback. This makes it difficult for students to share their skills effectively and improve where required.

A web-based system is needed to announce, track, and manage workshops, ensuring better participation, resource use, and feedback collection

Solution overview

The proposed solution is a web-based platform which students require. A single, easy-to-use online platform that takes the chaos out of peer-led workshops and replaces it with something that just works. Right now, organizing a skill-sharing session means juggling group chats, word-of-mouth, and hoping people show up. Our platform changes that. Students will be able to browse upcoming workshops, post their own sessions, and sign up, all from one place, without the back-and-forth.

One of the things we're most excited about is the automated reminder system. Low attendance is often less about disinterest and more about people simply forgetting. So, the platform will send email and SMS reminders as a workshop gets closer, giving participants a helpful nudge at just the right time. Behind the scenes, we're using Flask for the backend and a relational database to keep everything secure and well-organized. The interface itself is built with clean, responsive HTML5 and CSS3, so it's straightforward to navigate whether you're a first-year or a postgrad.

From the moment a workshop is scheduled to the point where feedback is collected, the platform handles the full journey. And throughout all of it, student data is protected in line with POPIA, because trust matters as much as functionality.

Key benefits and Functionality

This system aims to tackle the current issue of poorly organized workshops while also fostering student involvement, collaboration, and skill enhancement in an effective and lasting way.

Key Benefits

1. **Centralized Management and Visibility**
   The platform will provide a single, structured environment for publishing, managing, and monitoring all peer-led workshops. This eliminates reliance on fragmented communication channels and significantly improves accessibility and transparency.

2. **Improved Scheduling and Conflict Reduction**
   Time and venue conflicts will be reduced by the system's implementation of organize d scheduling elements. Better planning and coordination will be possible since students will be able to see future works earlier.

3. **Enhanced Participation Tracking**
   Accurate digital records of registrations and attendance will be stored by the system. Performance tracking, reporting, and identification of active participants and workshops host are made possible by this organized data management.

4. **Quality Assurance Through Feedback Mechanisms**
   Incorporating feedback and rating features will allow attendees to rate the workshops. This encourages responsibility, supports continuous growth, and guarantees that next classes fulfill student expectations and learning goals.

5. **Data-Driven Decision Making**
   Administrative dashboards and analytical tools will offer valuable insight into participation patterns, commonly discussed topics, and overall engagement levels. The information gathered can assist in making well-informed decisions when planning and improving future student development initiatives.

6. **Secure and Compliant Data Handling**
   In compliance with the principles of the Protection of Personal Information Act (POPIA) The system will employ secure user authentication procedures, store passwords using encryption techniques, and adhere to good data management standards. By taking these precautions, confidentiality will be preserved and personal data will be treated securely and responsibly.

7. **Automatic notification and reminder system**
   To inform users of upcoming workshops, the platform will integrate automated e-mail and SMS reminders. This feature increases overall engagement, lowers absenteeism, and promotes attendance consistency. Peer-led learning efforts are more effective when participants receive timely reminders to be informed and be prepared.

Core functionalities

These are the primary functional components that will be included in our application:

1. Safe user authentication and registration
2. Access control based on roles (administration and student roles)
3. Development, modification, and removal of workshops
4. Browsing and filtering in the workshop
5. Participant registration with capacity limits
6. Automated confirmation of registration

7. The ability to record attendance and the workshop
8. Submission of rating of previous workshop feedback
9. An administrative dashboard for oversight and monitoring
10. Analytical and reporting for participation data

Having all these features in the system will guarantee that the system is user friendly, safe, and in line with the institution's goal for students' growth and teamwork

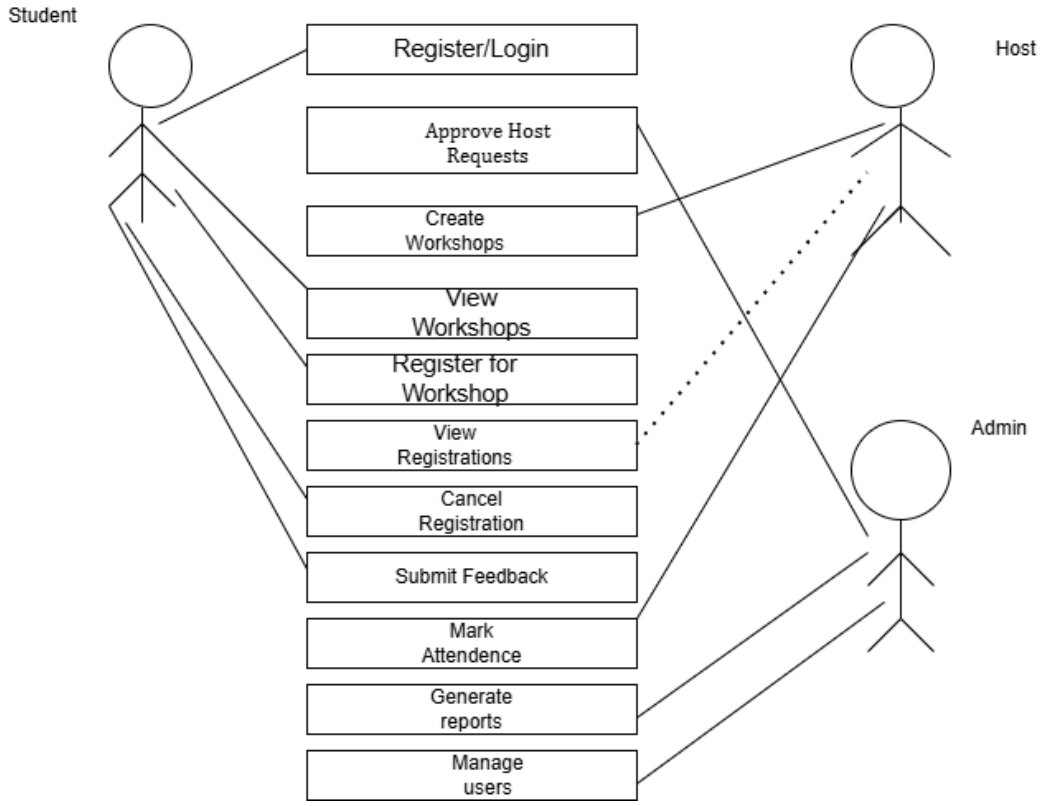
Milestones

Phase	Objective / Activities	Deliverables	Timeline
Research	• Analyze the current state of student-led workshops • Conduct surveys or interviews with students and organizers • Review similar scheduling tools and best practices	Research report, problem analysis summary	Week 1
Requirements Analysis	• Define functional requirements • Define non-functional requirements • Develop initial user stories and backlog	Requirements specification document, prioritized backlog of user stories	Week 2
Design	• Plan system architecture, database schema, and user interface • Develop prototypes for workshop scheduling features • Ensure modular and maintainable code structure	Design diagrams, wireframes, functional prototype	Week 3
Development	• Implement core features • Integrate frontend and backend • Conduct iterative testing and code reviews within development cycles	Working software increments demonstrating core functionality	Week 4
Testing	• Conduct unit, integration, and user acceptance testing • Identify and fix bugs, refine features • Validate security and compliance with POPIA • Perform performance and usability improvements	Test reports, bug logs, updated system ready for deployment	Week 5
Deployment		Deployed web-based solution,	Week 6

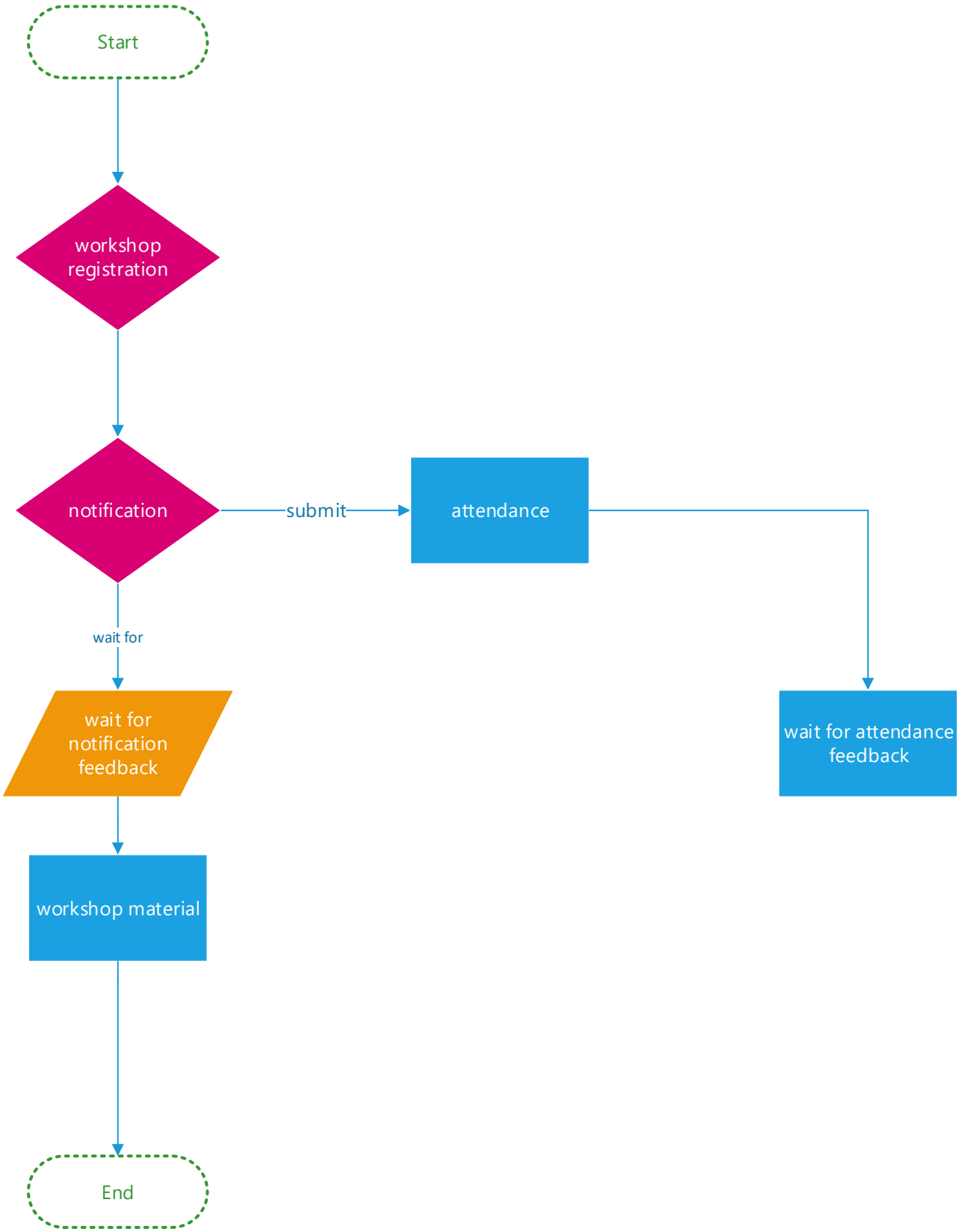
	• Deploy application to cloud hosting platform • Provide user access and documentation • Train student organizers on system usage	user manuals, deployment report	
Evaluation	• Collect feedback from users and stakeholders • Compare system performance against requirements • Identify lessons learned and recommendations for future improvements	Evaluation report, recommendations for future iterations	Week 7

Diagrams

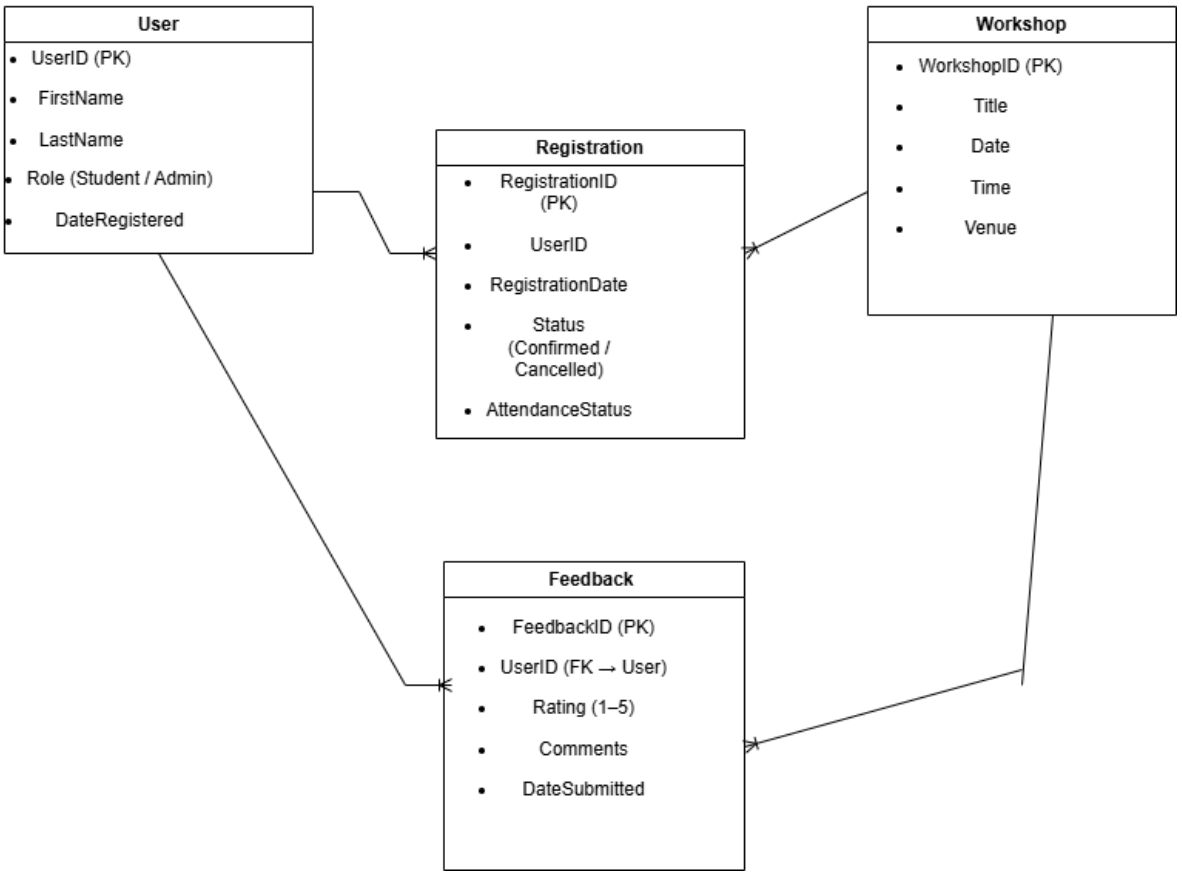
USE CASE DIAGRAM

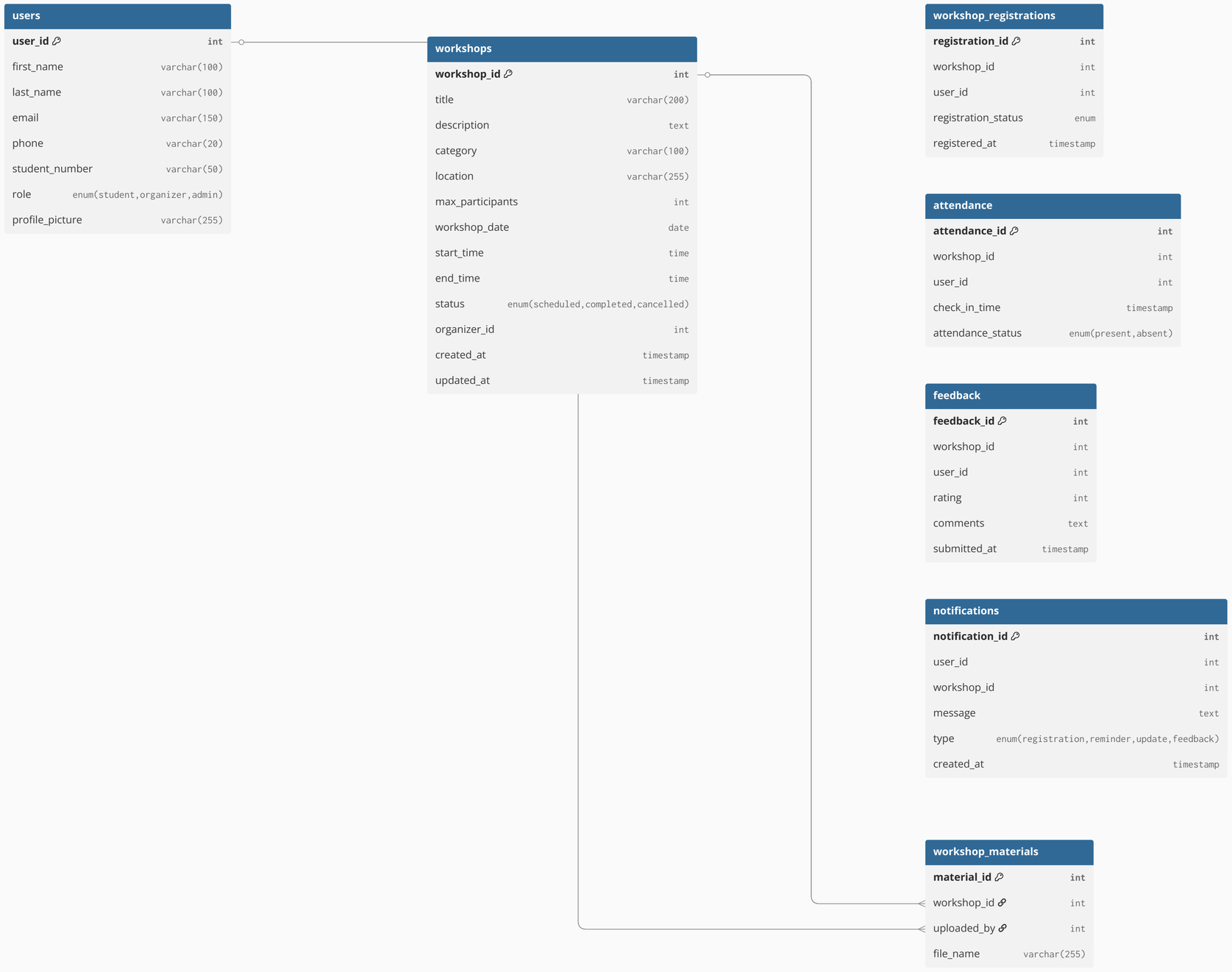


workshops user flow



ERD DIAGRAM


User
UserID (PK)
FirstName
LastName
Role (Student / Admin)
DateRegistered
Registration
RegistrationID (PK)
UserID
RegistrationDate
Status (Confirmed / Cancelled)
AttendanceStatus
Workshop
WorkshopID (PK)
Title
Date
Time
Venue
Feedback
FeedbackID (PK)
UserID (FK → User)
Rating (1–5)
Comments
DateSubmitted



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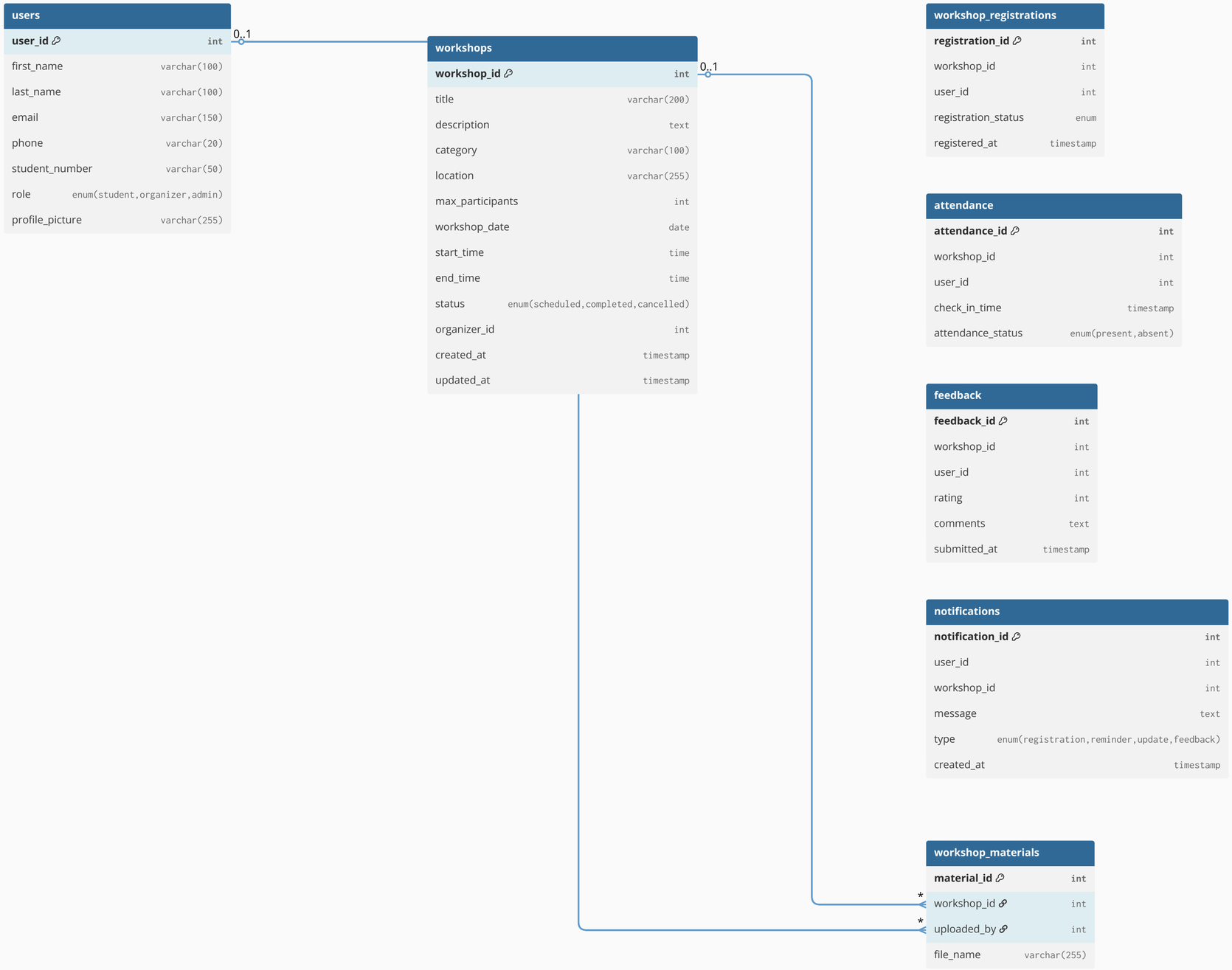

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Student Workshop System

Login

Email

Password

Forgot Password?

Login

Don't have an account? **Register**

Student Workshop System

Welcome, Alex!

- Dashboard
- My Workshops
- Profile
- Logout

Dashboard

My Workshops

My Created Workshops | Workshops I Joined

Workshop Title	Date & Time	Location	Participants	Status	Actions
Web Development Basics	Apr 28, 2024 - 3:00 PM	Room 101, CS Building	12	Scheduled	Edit Cancel
Creative Writing Workshop	Apr 30, 2024 - 2:00 PM	Library, Room 205	10	Pending	Edit Cancel
Data Analysis with Python	May 2, 2024 - 4:00 PM	Lab 3, IT Department	14	Scheduled	Edit Cancel

1 2

My Workshops

My Created Workshops | Workshops I Joined

Workshop Title	Date & Time	Location	Organizer	Status	Actions
Mobile App Development	Apr 26, 2024 - 1:00 PM	Room 102, CS Building	John S.	Registered	Unregister
Graphic Design Essentials	May 1, 2024 - 10:00 AM	Art Studio, Room 12	Emma L.	Waitlisted	Cancel Request
Machine Learning 101	May 5, 2024 - 3:00 PM	Room 305, CS Building	Michael K.	Closed	View Details

Student Workshop System

Alex

Dashboard

My Workshops

My Workshops

Manage and view your upcoming, attended, and organized workshops.

Upcoming Attended Organized

Upcoming Workshops

Web Development Basics

April 28 3:00 PM

Details Cancel

12 total Alex Doe

Creative Writing Workshop

April 30 2:00 PM

Details Cancel

10 total Emma L.

Data Analysis with Python

May 2 4:00 PM

Details Cancel

14 total John S.